

Software Development Multi-Track Certification Program

Final Project Presentation - Software Testing Track

Automated & Manual Testing of SauceDemo E-Commerce Platform

Candidate Name: Priyanki Panchal

Track: Software Testing

Academic Collaboration: BIT Baroda in academic association with IIT Patna - Vishlesan I-Hub

GitHub Repository: <https://github.com/priyankipanchal/09-Priyanki-SauceDemoTesting>



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Problem Statement & Test Architecture

Problem Statement:

- E-commerce systems require flawless execution of authentication, catalog operations, and checkout transactions to ensure data consistency and customer trust.
- The goal is to design an automated and manual QA verification pipeline to identify regressions, validate business logic, and verify edge-case behaviors.

Testing Architecture & Technology Stack:

- **Automation Engine:** Java with Selenium WebDriver
- **Build Tool:** Apache Maven (pom.xml dependency management)
- **Wait Strategy:** Explicit Dynamic Synchronization (WebDriverWait) to prevent test flakiness
- **Environment & Tools:** Windows 11, Google Chrome Browser, VS Code IDE
- **Version Control:** Incremental Git history with 15 commits on GitHub



Test Scope, Execution & Summary Metrics

Covered Modules: User Authentication, Product Catalog & Sorting, Cart Persistence, and Multi-Step Checkout Flow.

Metric / Deliverable	Count / Status	Evaluation Details
Total Documented Test Cases	20	Structured across 3 distinct functional modules in the test matrix
Automated Test Cases	5	Fully executable Java + Selenium WebDriver scripts
Manual Test Cases	15	Form validation, boundary cases, and layout verification
Passed Test Cases	17	100% of core transactional and checkout paths passed
Defects Logged (Bugs)	3	Documented with reproduction steps, severity, and screenshots
Pass Percentage	85%	Exceeds the mandatory 60% functionality pass floor



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Defect Analysis, Challenges & Reflection

Defect Findings (Bug Reports):

- **BUG_01:** Locked-out error message layout misalignment on narrow mobile viewports (Severity: Low).
- **BUG_02:** Missing explicit visual validation highlight on empty Postal Code field (Severity: Medium).
- **BUG_03:** Incorrect product thumbnail images displayed across catalog for Problem User (Severity: Medium).

Key Challenges & Resolutions:

- Resolved element lookup race conditions using dynamic explicit waits instead of thread sleeps.
- Structured project code incrementally across 15 version-controlled commits to demonstrate full development workflow.

Conclusion: All critical workflows function correctly end-to-end; documentation and deliverables align fully with certification criteria.